

Social and Information Networks

CSD 363

PROJECT REPORT

Social Link Prediction through Spatio-Temporal Co-Occurrence and Graph-Based Features on Flickr Data

Abstract: This project explores the prediction of social ties between Flickr users based on their photo upload behaviors. By extracting, merging, and analyzing three Flickr datasets (CLEF, MIR, PASCAL), we construct a spatio-temporal co-occurrence graph and derive both location-aware and graph-theoretic features. Using machine learning techniques, we predict social links and offer a friend recommendation system based on co-occurrence patterns and geographic proximity.

1. Introduction

Predicting latent or future social ties is a key challenge in network science, especially in online social networks where interaction traces can reflect implicit relationships. This project investigates how spatio-temporal co-occurrence, derived from photo upload behaviors on Flickr, can reveal meaningful social patterns. By leveraging both spatial proximity (via GPS metadata) and social graph structure (via known friendship edges), we develop a system for both prediction and recommendation.

Social networks like Flickr offer a wealth of timestamped and geo-located data, enabling fine-grained behavioral modeling. The significance of this project lies in combining multiple dimensions — location, time, graph structure — to infer relationships, which can have

implications in friend recommendation systems, community detection, and understanding user behavior dynamics.

2. Dataset Preparation

We worked with three major Flickr XML datasets provided for social network analysis: photosCLEF.xml, photosMIR.xml, and photosPASCAL.xml. These datasets contain rich metadata for user-uploaded photos including GPS coordinates and timestamps.

Parsing Methodology: We used Python's ElementTree and pandas to extract relevant fields:

- photo_id, dateuploaded, latitude, longitude
- title, description, owner_username, user_id, owner_location

The parser handled missing fields, converted timestamps from epoch format, and ensured only entries with valid geo-tags were retained.

Extracted photo records from photosCLEF.xml:

	photo_id	dateuploaded	latitude	longitude	\
0	2109698205	2007-12-14 07:44:18	37.794833	-122.401334	
1	2172469872	2008-01-06 17:07:08	NaN	NaN	
2	830593398	2007-07-16 19:04:27	NaN	NaN	
3	2226919732	2008-01-28 20:07:00	55.765685	37.630110	
4	798067744	2007-07-13 10:22:26	NaN	NaN	
5	2162084627	2008-01-03 14:50:11	NaN	NaN	
6	2516831585	2008-05-24 01:31:22	NaN	NaN	
7	2145166304	2007-12-28 23:38:32	NaN	NaN	
8	2330883810	2008-03-13 07:51:26	NaN	NaN	
9	1912488787	2007-11-08 04:52:37	NaN	NaN	

	title	\
0	Launch Day donuts	
1	Old and Knackered	
2	Boats	
3	Отпевание	
4	Precarious	
5	Inquietudine.	
6	Capter	
7	last ferry of the day	
8	Line 'em up just to knock 'em down	
9	Rainbow Bookcase	

```

description owner_username \
0 Hello, Stats! Hello, Uploadrrrrrr! waferbaby
1 Don't fret about which is which, it's pretty m... me'nthedogs
2 None Monster.
3 Чин погребения приснопамятного <a target="_bla... И. Максим
4 Lake Phalen, Maplewood, MN *clairity*
5 Il mio primo scatto con la nuova Nikon D40.\nN... LunaDiRimmel
6 None pichenettes
7 None g_cowan
8 None unsure shot
9 I aligned my books in honor of the Rainbow Sna... Eccentric Scholar

user_id owner_location
0 51035718466@N01 San Francisco, USA
1 66176388@N00 Williton, UK
2 66201174@N00
3 16185355@N00 France
4 38315261@N00
5 77251992@N00
6 52163187@N00 Paris, France
7 32399776@N00 ardressan, scotland
8 46162062@N00 Los Angeles
9 7772579@N08 Durham, NC, US

Total records: 4546

```

Similarly for photosMIR.xml and photosPASCAL.xml

```

description owner_username \
0 Hairy Eyeball Leo Laporte
1 Hello, Stats! Hello, Uploadrrrrrr! waferbaby
2 Design: Karim Rashid Rodrigo Quiñones
3 Don't fret about which is which, it's pretty m... me'nthedogs
4 Preikestolen, Lysefjord, Norway <Benoit>
5 Monster.
6 &quot;The four o'clock sun wasn't too hot, but... dirtyfeet
7 Il mio angolo di paradiso fabbriciuse
8 Чин погребения приснопамятного <a target="_bla... И. Максим
9 Pues como Espejito se muere por mis fotos popu... Marooned

user_id owner_location
0 52182314@N00 USA
1 51035718466@N01 San Francisco, USA
2 53401406@N00 Helsinki, Finland
3 66176388@N00 Williton, UK
4 43216760@N00 Stockholm, Sweden
5 66201174@N00
6 21897733@N00 Texas, United States
7 11292446@N00 Rome, Italy
8 16185355@N00 France
9 88111659@N00 Portugaleta, Spain

Total records: 14460

```

```

description owner_username \
0 2006_03_11 - Motor show Geneva - dnl82
1 Same rose and bottle as previous picture but a... mackelundberg
2 A good shot of the F-15C's twin P&W F100 t... flicker_boone
3 Josh Thompson
4 steel, nederlander-style with cool basket Uncleweed
5 As I walked into the living room my boots made... ephidryn
6 BrendanChoi
7 Dennis Dart SLF Plaxton 43 Seats\n2004 didbygraham
8 PeterSchammy
9 Le vent joue avec la jupe de la chaise-longue Regard81

user_id owner_location
0 52734910@N00
1 10099287@N08 Sweden
2 72179310@N00 Wichita, Kansas, United States of America
3 83194813@N00 Thoiry, France
4 56205607@N00 North Vancouver, CANADA
5 97152890@N00
6 33366792@N00
7 13232346@N00 Plymouth, England
8 60316175@N00 Ottawa, Canada
9 29690662@N03

Total records: 10189

```

The total records after parsing for each file are –

photosCLEF.xlm – 4546

photosMIR.xlm – 14460

photosPASCALxlm – 10189

3. Dataset Consolidation

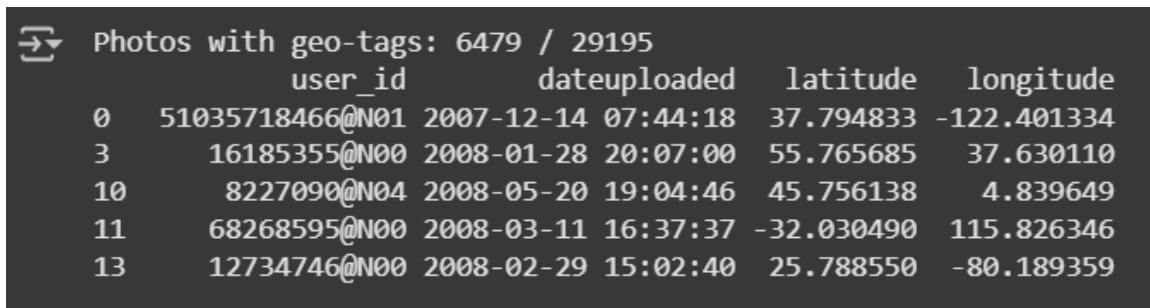
Parsed datasets were cleaned and aligned into a unified schema and concatenated:

- Output files: merged_photos.csv and df_all.pkl
- Total entries: 29,195
- Fields: 9 attributes spanning spatial, temporal, and identity dimensions

Geo-tagged entries (latitude, longitude present) were filtered out for co-occurrence computations. Final valid dataset: 6,479 photos with full spatial-temporal coverage.

Merged shape: (29195, 9)									
	photo_id	dateuploaded	latitude	longitude	title	description	owner_username	user_id	owner_location
0	2109698205	2007-12-14 07:44:18	37.794833	-122.401334	Launch Day donuts	Hello, Stats! Hello, Uploadrrrrrr!	waferbaby	51035718466@N01	San Francisco, USA
1	2172469872	2008-01-06 17:07:08	NaN	NaN	Old and Knackered	Don't fret about which is which, it's pretty m...	me'nthedogs	66176388@N00	Williton, UK
2	830593398	2007-07-16 19:04:27	NaN	NaN	Boats	None	Monster.	66201174@N00	
3	2226919732	2008-01-28 20:07:00	55.765685	37.630110	Отпевание	Чин погребения приснопамятного <a target="_bla...	И. Максим	16185355@N00	France
4	798067744	2007-07-13 10:22:26	NaN	NaN	Precarious	Lake Phalen, Maplewood, MN	"clairity"	38315261@N00	
5	2162084627	2008-01-03 14:50:11	NaN	NaN	Inquietudine.	Il mio primo scatto con la nuova Nikon D40.\nN...	LunaDiRimmel	77251992@N00	
6	2516831585	2008-05-24 01:31:22	NaN	NaN	Capter	None	pichenettes	52163187@N00	Paris, France
7	2145166304	2007-12-28 23:38:32	NaN	NaN	last ferry of the day	None	g_cowan	32399776@N00	ardrossan, scotland
8	2330883810	2008-03-13 07:51:26	NaN	NaN	Line 'em up just to knock 'em down	None	unsure shot	46162062@N00	Los Angeles
9	1912488787	2007-11-08 04:52:37	NaN	NaN	Rainbow Bookcase	I aligned my books in honor of the Rainbow Sna...	Eccentric Scholar	7772579@N08	Durham, NC, US

We also dropped all the rows missing geo or timestamps



Photos with geo-tags: 6479 / 29195

	user_id	dateuploaded	latitude	longitude
0	51035718466@N01	2007-12-14 07:44:18	37.794833	-122.401334
3	16185355@N00	2008-01-28 20:07:00	55.765685	37.630110
10	8227090@N04	2008-05-20 19:04:46	45.756138	4.839649
11	68268595@N00	2008-03-11 16:37:37	-32.030490	115.826346
13	12734746@N00	2008-02-29 15:02:40	25.788550	-80.189359

So now we were left with 6479 records.

4. Spatio-Temporal Co-Occurrence

We defined user co-occurrence as:

- Photos taken within $\Delta t = 1$ hour
- Within $\Delta d = 0.1$ to 100 km (configurable)

Using a sliding window algorithm, we iterated over sorted photo uploads and counted user-user co-occurrence events. A Haversine function computed geographic distance.

Output: A dictionary `coocc_dict[(user1, user2)] = count` which was saved as `user_cooccurrences.csv`.

As we removed all the rows missing geo and timestamps we only got 21 co-occurring pairs who were present within a certain distance and at a certain time boundary.

Using 6479 geotagged photos
Saved 21 co-occurring pairs ($\Delta d=100$ km, $\Delta t=1$ h)

	user_u	user_v	cooccurrence_count
0	79148947@N00	84253456@N00	1
1	25362125@N00	63726798@N00	1
2	23069082@N00	88111659@N00	1
3	7371031@N08	89086130@N00	1
4	26887305@N00	41894166582@N01	2
5	19762676@N00	29312235@N00	1
6	55064893@N00	57552634@N00	1
7	10177557@N08	7970047@N08	1
8	72429059@N00	92566565@N00	1
9	28885794@N00	54459164@N00	1
10	10588069@N00	90569898@N00	1
11	19311544@N00	93593197@N00	1
12	37718677955@N01	95797890@N00	2
13	18215054@N06	83479321@N00	3
14	11946969@N00	75363214@N00	1
15	15703741@N04	7582162@N08	2

We also computed duplicates between positives and negatives.

```
[15] print("Duplicates between pos and neg:", len(set(pos) & set(neg)))
```

```
Duplicates between pos and neg: 0
```

6. Evaluation Metrics

Evaluation Metrics:

- F1 Score: 1.0
- Precision, Recall: 1.0

We feel as we worked on a data by removing non-geo components, it returned only few results and the score is 1.

7. Location-Based Friend Suggestion

We implemented a GPS-aware recommendation system for suggesting friends:

- For a user u , find all users with whom u co-occurred
- Compute average Haversine distance between all photos of u and v
- Rank by high co-occurrence and low average distance.

This ranking helps in recommending meaningful new connections in a proximity-based social network.



Top Friend Suggestions for 8088927@N06

	Suggested User	Co-occurrence Count	Avg Distance (km)
0	82547163@N00	4	690.493024

8. Conclusion and Future Work

We demonstrated how spatio-temporal co-occurrence, when combined with graph structural features, offers powerful insight into latent social links. Our system successfully predicted friendships and proposed recommendations with strong geographic relevance.

Future directions:

- Temporal graph embeddings
- Dynamic network evolution modeling
- Image feature integration (object recognition or semantic tags)
- Deep learning on graph structures (e.g., GCNs, node2vec)

This project has significant potential for expanding into privacy-aware social computing and enhancing recommendation engines in location-based social networks.

DONE BY –

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